

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 07

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Waste products that are the raw materials for other processes	1
01.2	• Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) • The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂	2
01.3	• Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
02.1	• Anchovy r-selected; blue whale k-selected • r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) • k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos)	3
02.2	• Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly • An increase in death rate above the low birth rate causes a population crash • Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops	3
02.3	• New population = starting population + births + immigrants - deaths - emigrants	1

Question	Marking guidance	Marks
02.4	• The greatest rate of exploitation (harvest) possible without causing long-term population decline • Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) • At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	
03.1	• Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity • Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes • Judgement depends on species, market and site	4
03.2	• Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation • Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow • After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg	3
03.3	• About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection • Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose • Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
04.1	• The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out • Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost	2
04.2	<i>Any 3 from:</i> • Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth • The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding • Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients • Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted	3
04.3	• Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances • Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving); specialised flowers reduce pollen being wasted on other species	2

Question	Marking guidance	Marks
04.4	• Dispersal is not haphazard: animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive • Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling • Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
05.1	• $1000 / 100\,000 \times 100 = 1\%$	1
05.2	• Some light is reflected by the leaves • Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed • Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration)	3
05.3	• Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) • After four levels too little energy remains to support a further population	2
05.4	• Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) • Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat	2
05.5	• Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) • So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
06.1	• General physiological solvent: most biological reactions take place between substances dissolved in water • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats	4
06.2	• From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) • Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ • The ozone layer reduced UV reaching the surface so organisms could survive out of the water • Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out	4
06.3	• Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels • This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
07.1	• B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings • B is warmer / longer thermal growing season because south-facing slopes receive more insolation; valley floors are frost pockets where cold dense air collects • But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed	3
07.2	<i>Any 3 from:</i> • Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) • Woven plastic cloches / fleece over rows to retain warm air near the ground • Burning oil / gas or using large fans in orchards to prevent frost damage to blossom • Site selection (aspect, avoiding frost pockets); heated buildings and shelter for livestock	3
07.3	• Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) • Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth	2
07.4	• Long days increase milk production and egg laying; poultry grow best with short days; sheep mate as days shorten in autumn • Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	
08.1	• Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable • If it cannot be adjusted the percentage error should be recorded so the results can be corrected	2
08.2	<i>Any 3 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (eg chewed hazel nuts): presence of species such as dormice • Nests / burrows: presence and breeding • Tracks / footprints: presence, movements • Territorial marks (scratching posts): presence and territory	3
08.3	• Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen	2

Question	Marking guidance	Marks
08.4	<i>Any 3 from:</i> • Marked individuals mix freely with the rest of the population • Marks do not fall off or wear away, and marking does not affect survival • The population does not change during the study (no births, deaths or migration) • All individuals have an equal chance of being caught / animals are mobile and can be caught	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
09.1	• $600 + (90 + 150) - (220 + 80) = 540$ • Stock fell by 60 so the catch exceeds the MSY / unsustainable	2
09.2	• $A + G - M = 90 + 150 - 80 = 160$	1
09.3	• Catch is landed and weighed at ports so it is recorded • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years	3
09.4	• CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments • Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally • Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
10.2	• Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25